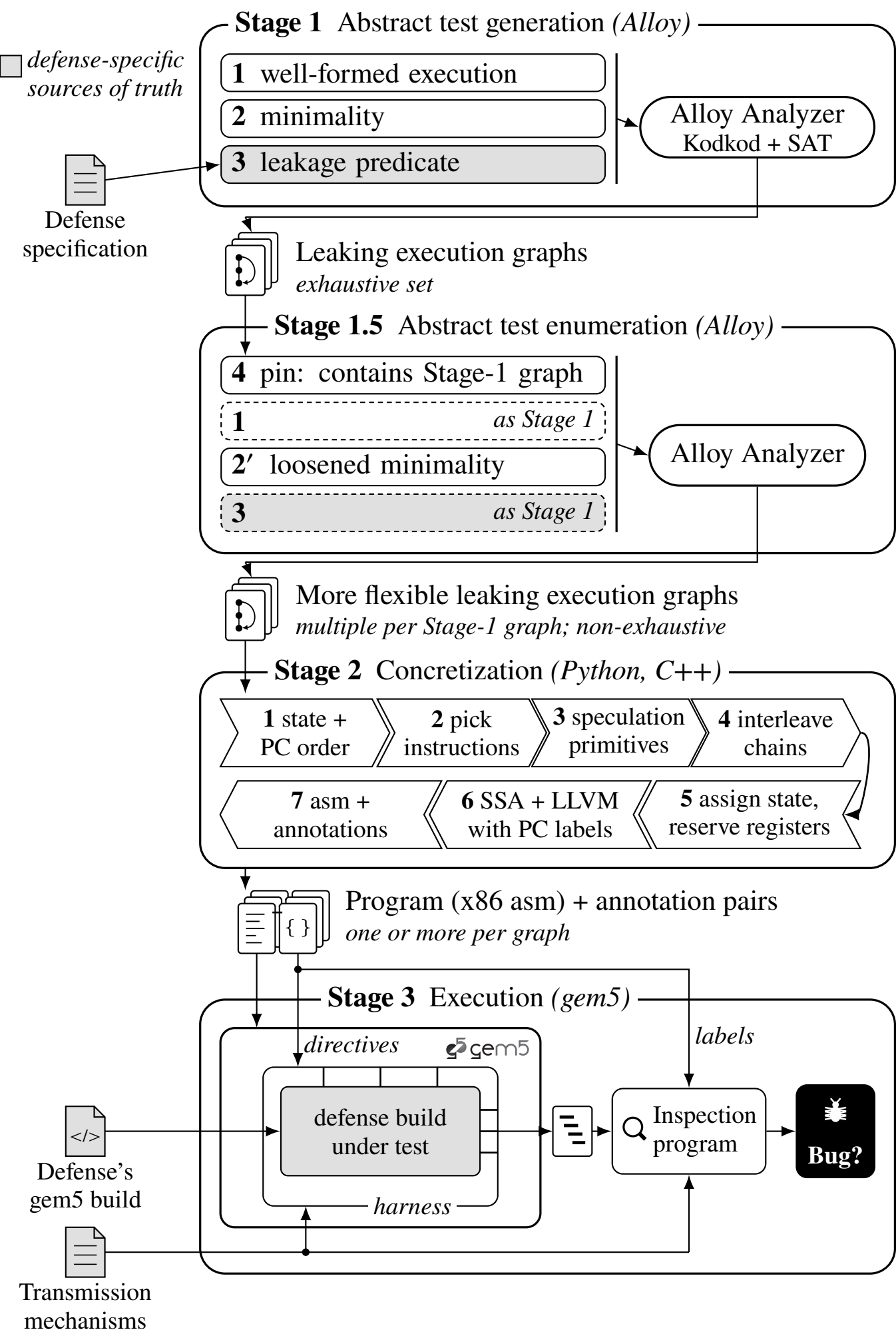

Stage 1 Abstract test generation (Alloy)
defense-specific sources of truth
1 well-formed execution
2 minimality
3 leakage predicate
Alloy Analyzer
Kodkod + SAT
Defense specification
Leaking execution graphs
exhaustive set
Stage 1.5 Abstract test enumeration (Alloy)
4 pin: contains Stage-1 graph
1 as Stage 1
2′ loosened minimality
3 as Stage 1
Alloy Analyzer
More flexible leaking execution graphs
multiple per Stage-1 graph; non-exhaustive
Stage 2 Concretization (Python, C++)
1 state + PC order
2 pick instructions
3 speculation primitives
4 interleave chains
5 assign state, reserve registers
6 SSA + LLVM with PC labels
7 asm + annotations
Program (x86 asm) + annotation pairs
one or more per graph
Stage 3 Execution (gem5)
directives
gem5
defense build under test
harness
labels
Inspection program
Bug?
Defense's gem5 build
Transmission mechanisms